

Data-Aided Radar Sensing using 5G NR Systems: Performance Limits and Implementation

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Introduction

Integrated sensing and communication (ISAC) is a key beyond-5G/6G feature in which the same radio infrastructure supports communication and environmental sensing. Among the various physical layer waveforms being considered for ISAC, Orthogonal Frequency Division Multiplexing (OFDM) is the de-facto waveform for 5G New Radio (5G NR). Prior works have already demonstrated efficient Delay–Doppler processing [7], multiband operation [1], waveform design approaches [4] and MIMO extensions [3, 5] for OFDM-ISAC systems. In addition, recent work has demonstrated the use of 5G control channels such as the Positioning Reference Signal (PRS) [3], Demodulation Reference Signal (DMRS), and Channel State Information Reference Signal (CSI-RS) [9, 13]. However, most of the prior works consider OFDM cyclic prefix (CP)-limited sensing regimes, wherein the Range-Doppler Map (RDM) is estimated post-CP removal, using matched filtering with known reference (for bistatic radar sensing) or data (for monostatic radar sensing) signals. Recently, Xu *et al.* [14] have investigated methods to sense targets beyond the range imposed by the CP duration, but assume low velocity targets such that radar signal processing is feasible in the OFDM time-frequency grid.

Problem Statement

The focus of this proposal is to enable radar sensing using 5G NR signals for long range and high velocity targets. To motivate the problem, we consider a monostatic OFDM sensing scenario where the RDM is obtained by the standard method of correlating the received signal with a known delay-shifted and Doppler-rotated version of the transmit reference. Let *reference-only* (RO) sensing denote the case where only reference signal resource elements are used to generate the transmit reference signal, and let *data + reference signal* (DR) sensing denote the case where both reference signal and data-bearing elements are used to generate the transmit reference signal. A denser DR reference is desirable because it provides stronger coherent gain and a larger unambiguous range–velocity support [3]. If reference signal spacings in frequency and time are N_f subcarriers and N_t OFDM symbols, then RO reduces the unambiguous sensing support by factors $1/N_f$ in range and $1/N_t$ in Doppler relative to DR. Sparse pilot patterns therefore induces *aliasing* in the delay–Doppler plane and creates deterministic ambiguity replicas across the RDM.

These replicas appear as ghost targets and can trigger bins outside the true target support, as shown in Fig. 1. If $A = N_f N_t$ denotes the aliasing factor, K is the number of true targets, and $N_{\text{total}} = N_{\text{range}} N_{\text{dopp}}$ is the number of Range-Doppler (RD) cells, then the ghost-induced false-alarm probability scales approximately as

$$P_{\text{FA}}^{\text{ghost}} \approx \frac{K(A - 1)}{N_{\text{total}}}. \quad (1)$$

Intuitively, this is because for each true target detected, there will be $(A - 1)$ ghost targets due to aliasing. This captures the dominant dependence on pilot sparsity and target count, even though it does not attempt a full exhaustive characterization of other sources such as random data-induced sidelobes and the behavior of standard algorithms such as the *Cell Averaging - Constant False Alarm Rate* (CA-CFAR) detector. This issue is especially important because the CA-CFAR detector will fail to alleviate this issue, as its design assumes clutter and noise models that do not account for range and Doppler ambiguity.

Considering this observation, this proposal asks:

What are the performance limits of data-aided radar sensing of long range and high velocity targets using 5G NR signals, and how do we design efficient radar signal processing algorithms to detect and track such targets when leveraging high dimensional data signals?

For the purposes of this proposal, ‘data-aided’ means that we will use the entire 5G NR signal for radar sensing.

Proposal Description

System model

We consider a monostatic multi-antenna OFDM/5G NR sensing receiver. Let $X[k, n]$ denote the transmitted symbol on subcarrier k and OFDM symbol n , with subcarrier spacing Δf and symbol duration T_{sym} (including

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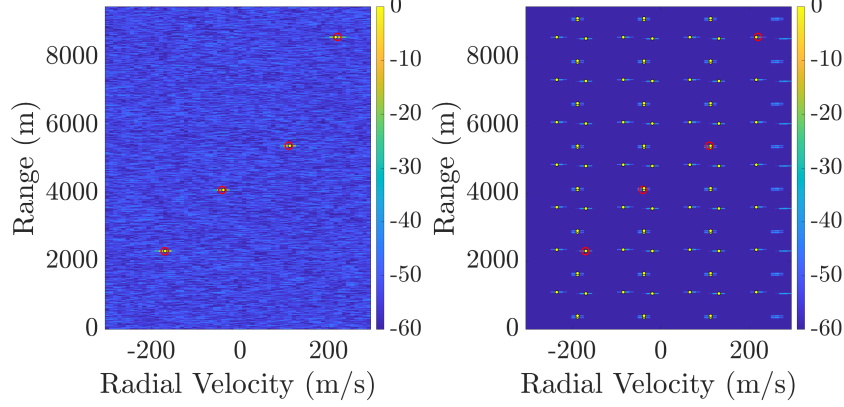


Figure 1: DR and RO range-Doppler maps under sparse pilot sampling. RO creates deterministic aliasing replicas that persist after CA-CFAR because the detector background is no longer noise-dominated.

cyclic prefix). The continuous-time received signal at antenna m can be modeled as

$$y_m(t) = \sum_{p=1}^P \alpha_p a_m(\theta_p) e^{j2\pi\nu_p t} \sum_{n=0}^{N_s-1} \sum_{k=0}^{N_c-1} X[k, n] e^{j2\pi k \Delta f (t - \tau_p - nT_{\text{sym}})} \text{rect}\left(\frac{t - \tau_p - nT_{\text{sym}}}{T_{\text{sym}}}\right) + w_m(t), \quad (2)$$

where α_p , τ_p , ν_p , θ_p , and $a_m(\theta_p)$ denote the channel coefficient, delay, Doppler, azimuth AoA, and m^{th} element of the steering vector corresponding to the p^{th} target. $w_m(t)$ denotes continuous-time additive noise. The number of subcarriers is N_c , the number of OFDM symbols per radar frame is N_s , and the rectangular window function is denoted by $\text{rect}(\cdot)$.

Let $\mathcal{R} \subset \mathcal{S}$ denote the index set of reference signal resource elements,² where $\mathcal{S} = \{0, \dots, N_c - 1\} \times \{0, \dots, N_s - 1\}$. Then, the data resource elements are indexed by $\mathcal{D} = \mathcal{S} \setminus \mathcal{R}$. To estimate the target range and Doppler, we construct a discrete-time reference waveform $s_{\text{ref}}[u]$ by applying OFDM modulation to the symbols in $\mathcal{R}$ while setting $\mathcal{D}$ to zero. Sampling the received signal $y_m(t)$ at interval T_s , we evaluate the discrete cross-ambiguity function over a grid of delay indices $l \in \{0, \dots, L - 1\}$ and Doppler hypotheses $d \in \{0, \dots, N_d - 1\}$ as

$$R_m[l, d] \triangleq \sum_{u=0}^{N_c-1} y_m[u] s_{\text{ref}}^*[u - l] e^{-j2\pi\nu_d^{(d)} u T_s}, \quad (3)$$

where L corresponds to the maximum delay of interest, and $\nu_d^{(d)}$ is the d^{th} Doppler hypothesis. Target parameters are extracted via peak detection on the range-Doppler power map $P_m[l, d] = |R_m[l, d]|^2$. In the DR scheme, successfully decoded data symbols in $\mathcal{D}$ can act as virtual pilots to reconstruct a full-frame reference $s_{\text{ref}}[u]$, significantly boosting processing gain.

Proposed Objectives and Methodology

Building upon this DR processing framework, evaluating the cross-ambiguity function in dense 5G NR grids for long-range and high-velocity targets introduces severe theoretical and computational bottlenecks. To address these, this one-year QIF proposal focuses on four key research objectives:

1. Theoretical Characterization of Ghost Targets and Aliasing: While the DR scheme mitigates deterministic aliasing compared to the RO scheme, long-range and high-velocity targets in 5G NR systems suffer from range and Doppler migration (i.e., targets straddling multiple delay/Doppler bins across the frame). We will develop an analytical framework to characterize the bounds of the false alarm probability under multi-target interference and migration effects. By modeling the statistical distribution of the ambiguity sidelobes in the DR scheme versus the deterministic replicas in the RO scheme, we will derive analytical approximations for the effective probability of false alarm (P_{FA}) and probability of detection (P_{D}), providing a theoretical baseline for 5G ISAC performance limits.

2. System-Level Performance under Diverse gNB Deployments: 5G NR systems operate in complex real-world geometries, meaning the radar link budget is heavily influenced by the deployment scenario. We will evaluate the conditions under which long-range targets can be reliably detected under different gNB heights (e.g., macro-cell

²For the purposes of this proposal, a ‘reference signal’ is comprised of the 5G NR synchronization signals (PSS/SSS), reference signals such as DMRS/CSI-RS/PRS/etc., and other control channels that are known/can be reliably decoded by the radar receiver.

vs. small-cell) and downtilts. Crucially, this objective will explicitly incorporate geometry-dependent *clutter models* (e.g., ground and building reflections). Because DR sensing relies on continuous data transmission, it is highly susceptible to near-field clutter masking far-field targets. We will establish the signal-to-interference-plus-noise ratio (SINR) bounds required to overcome geometry-specific clutter profiles.

3. Low-Complexity RDM Signal Processing: The brute-force evaluation of the discrete cross-ambiguity function $R_m[l, d]$ has a computational complexity of $\mathcal{O}(N_c \times L \times N_d)$, which is prohibitive for real-time tracking of high-velocity targets over large 5G bandwidths. To overcome this, we will develop low-complexity signal processing algorithms to estimate the RDM. We will investigate multi-stage processing pipelines leveraging fast-time and slow-time Fast Fourier Transforms (FFTs), decimation techniques, and sparse recovery algorithms (such as Orthogonal Matching Pursuit) that exploit the sparsity of the target scene. The goal is to achieve near-optimal DR sensing performance at a fraction of the computational cost of matched filtering.

4. Experimental Validation via USRP Proof-of-Concept: To transition the proposed theories and algorithms from simulation to practice, we will develop a hardware proof-of-concept using National Instruments (NI) USRP software-defined radios. We will establish a monostatic testbed operating at sub-6 GHz frequencies to capture over-the-air 5G-NR OFDM sensing signals. To accommodate the one-year project timeline, baseband data generation and high-bandwidth signal processing will be performed offline using MATLAB/Python. This will allow us to validate the theoretical false alarm bounds and the efficiency of the fast algorithms using real-world hardware impairments (e.g., phase noise, LO leakage, and timing offsets).

Execution Plan and Timeline

The proposed tasks are part of a multi-year effort to develop radar sensing capabilities in 5G NR to detect long-range and high-velocity targets. A timeline of the proposed research activities aligned with the initial 12-month period is provided below. The timeline is structured to ensure parallel progress on theoretical derivations and hardware setup, thereby minimizing the risk of bottlenecks.

Quarter	Milestones and Deliverables
Q1	Theory & Hardware Setup: Establish the mathematical clutter models for various gNB heights (Obj 2). Procure USRPs, configure antennas, and establish the offline loopback baseline (Obj 4).
Q2	Derivations & Algorithms: Derive theoretical bounds for migration-induced false alarms (Obj 1). Begin formulation of fast-time/slow-time FFT and sparse RDM processing algorithms (Obj 3).
Q3	Refinement & Data Collection: Finalize low-complexity algorithms and benchmark their FLOP counts against standard CA-CFAR. Conduct extensive over-the-air measurements with the USRPs in varied clutter environments (Obj 4).
Q4	Validation & Dissemination: Run offline processing on the captured hardware data. Validate theoretical bounds against experimental RDMs. Compile findings into a final report and target a high-impact IEEE journal or flagship conference (e.g., IEEE ICC/Globecom/ICASSP).

Team Qualifications and Experience

Student 1: Amit K. Yadav is currently a Master’s student at IIT Gandhinagar and previously worked as a 5G NR Protocol Developer at Jio Reliance Infocom Pvt. Ltd. from July 2022 to July 2025. At Jio, he contributed to multi-BWP support, RRM resource dimensioning, MOCN enablement, NRPPa transaction development, and UAC barring implementation, with validation across Real UE and Keysight-based test setups. This experience provides strong preparation for the proposed work on OFDM/5G NR sensing, pilot behavior, and robust signal processing under practical system constraints.

Student 2: Rupanjan Ghosh is currently a Junior Undergraduate in Electrical Engineering at IIT Gandhinagar with a strong foundation in signal processing, machine learning, electromagnetics, and control systems. He has worked on bistatic radar micro-Doppler analysis at IIT Bombay, deriving analytical expressions for moving-target signatures and comparing bistatic and monostatic configurations for target identification. His experience also includes solenoid design using mathematical modelling and FEMM-based simulation, along with ongoing work on hearing-aid feedback cancellation using adaptive filtering and sub-band decomposition. This background prepares him well for research in radar sensing, adaptive signal processing, electromagnetic modelling, and numerically validated system design.

Faculty Advisor: Raghunandan M. Rao is currently an Assistant Professor in the Department of Electrical Engineering at IIT Gandhinagar. Prior to this, he held full-time engineering and research positions at Amazon Lab126, Sunnyvale, CA (Jan 2022 - Nov 2024) and Samsung Research America, Plano, TX (Oct 2020 - Dec 2021) in the United States, where he worked on signal processing and use-case development for Ultra-wideband (UWB) and millimeter-wave (mmWave) radar-based positioning and sensing systems. He earned his PhD in Electrical Engineering from Virginia Tech in 2020, where he made research contributions to 4G/5G control channel vulnerabilities, link adaptation, and interference modeling for radar-cellular spectrum sharing scenarios.

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